

UNIVERSITY EXAMINATION BOARD
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
TEST EXAMINATION

Subject: Design and Analysis of Algorithms	Sub Code: CS-501	Max Marks: 70
Time / Duration: 3 Hours	Academic Year: 2026-2027	Semester: V Semester

General Instructions:

- 1. Answer all questions in Part A.
- 2. In Part B, answer either (a) or (b) from each question.
- 3. Assume suitable missing data if necessary.

PART A

SHORT BITS

1. Define Big-O notation. **[2 Marks]**
2. State the Master theorem. **[2 Marks]**

PART B

DESCRIPTIVE

11. (a) Explain Quick Sort partition algorithm. **[10 Marks]**
- [OR]**
- (b) Explain Merge Sort and analyze time complexity. **[10 Marks]**